ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

Pixel pitch = 0.005 mm, sensor width = 1280x0.005 = 6.4 mm. how to calculate the belt field of view.

Q1(b) [3 marks]

$$Sensitivity = 10mV/N$$

The ADC has 12 bits, so there are 4096 levels.

Q1(c) [3 marks]

Question 2 [10 marks]

Q2(a) [2 marks]

Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

Q2(c) [4 marks]

At rest, 50(π/2-θss) = 2.4525. This gives θss = 1.5218 rad = 87.19°, so error approximately 2.81°.

Q2(d) [2 marks]

The added gravity torque should cancel the load, so the steady-state error becomes zero.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differences are poor mainly because the encoder measures angle in degrees instead of radians, and because the velocity signal is always delayed by one full second.

Q3(b) [2 marks]

$$y[k] = 0.8818y[k-1] + 0.0591x_k + 0.0591x[k-1], DCgain1$$

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. The desired polynomial is $(s+10)^2 = s^2+20s+100$.

Q3(d) [2 marks]

The observer is better and the filter is safer in some cases.

Question 4 [10 marks]

Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

Q4(b) [2 marks]

The actual feedforward contribution is 0.75x8=6.

Q4(c) [4 marks]

$$k_1 = 18, k_2 = 6, K_I = 32$$

Q4(d) [2 marks]

At steady state, a stable closed loop with integral action drives $\dot{\xi}$=r-θ to zero; therefore r-θ=0 and the tracking error is zero, assuming no saturation or other implementation limits prevent convergence. A practical failure mode is integrator windup during actuator saturation; the standard remedy is anti-windup, such as back-calculation or conditional integration.

With integral action, a constant tracking error keeps accumulating in the integral state, so the controller can adjust its output until the error goes to zero, assuming the closed loop is stable and the actuator is not constrained. Under saturation the integral state can keep growing; anti-windup limits or back-calculates that state.